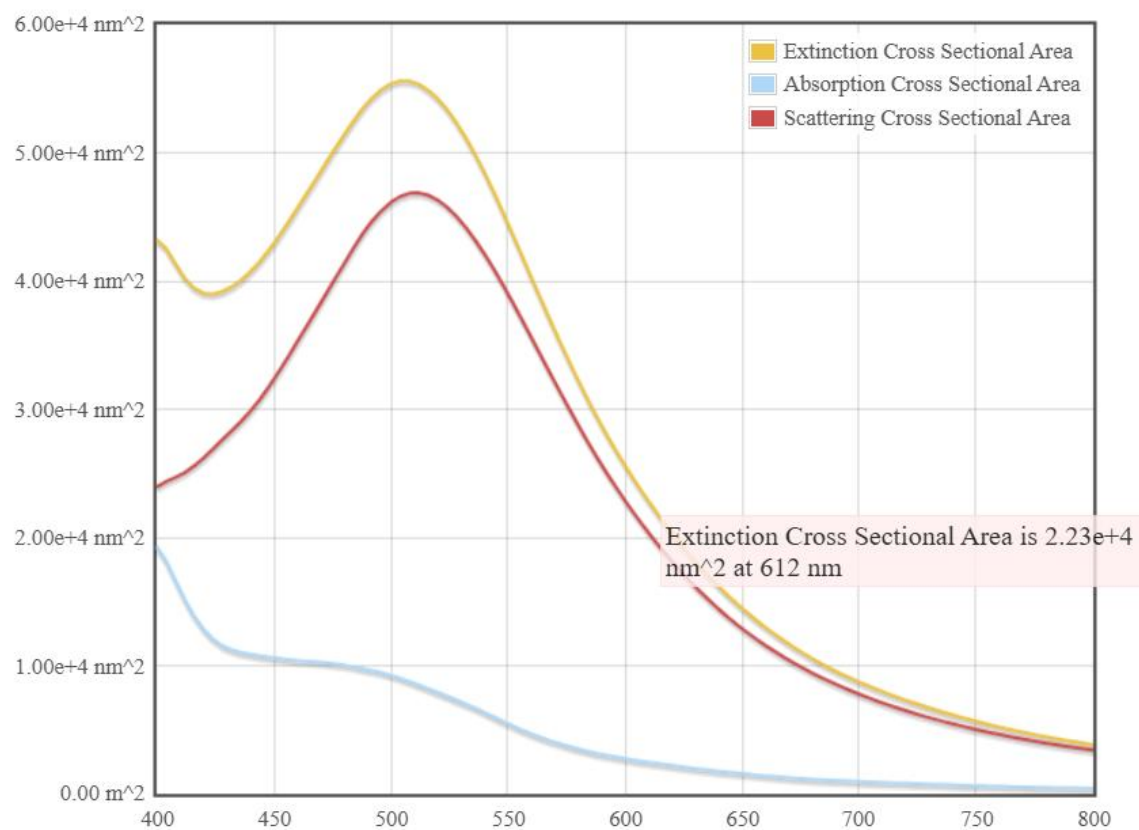


Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and
Core-Shell Nanoparticle using Mie-Plot tool

Output:

Scattering and absorption spectra for a core-shell nanoparticle.

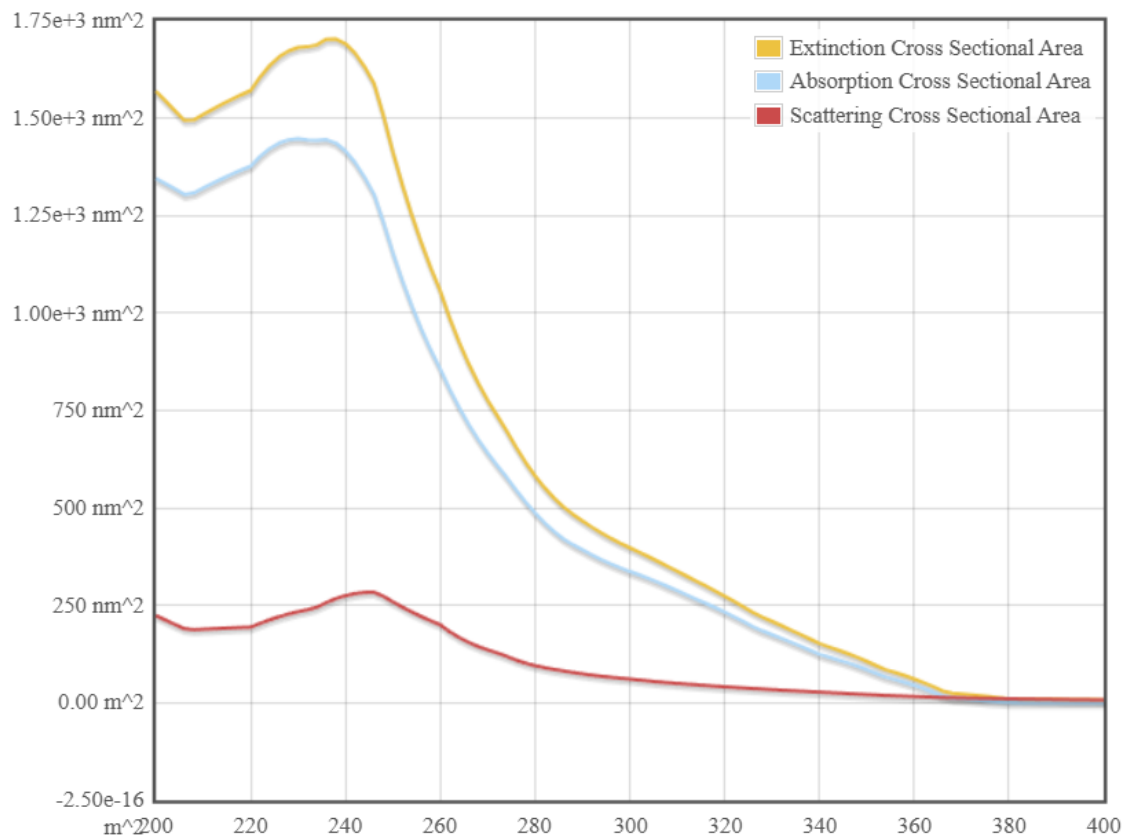
Core Type	Core RI	Core Diameter (nm)
Gold		50
Shell Type	Shell RI	Shell Thickness (nm)
Silver		25
Start λ (nm)	End λ (nm)	Medium RI
400	800	1.33



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

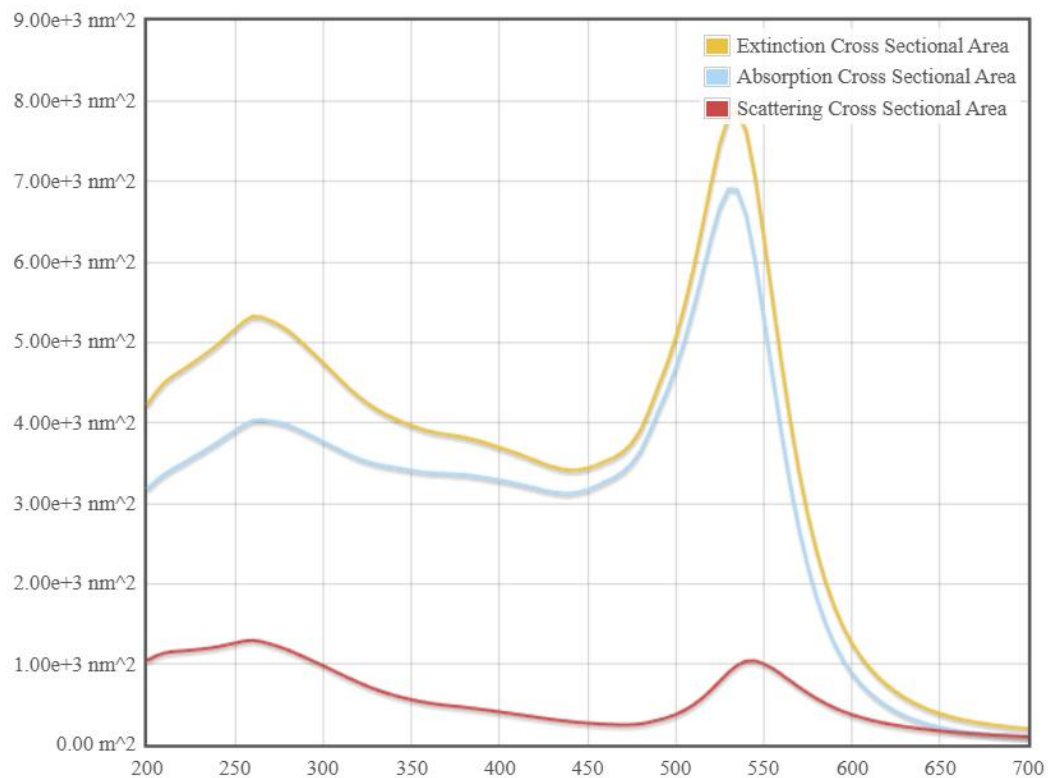
Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).

Core Type	Core RI	Core Diameter (nm)
TiO<sub>2</sub>		30
Shell Type	Shell RI	Shell Thickness (nm)
None		0
Start λ (nm)	End λ (nm)	Medium RI
200	400	1.33



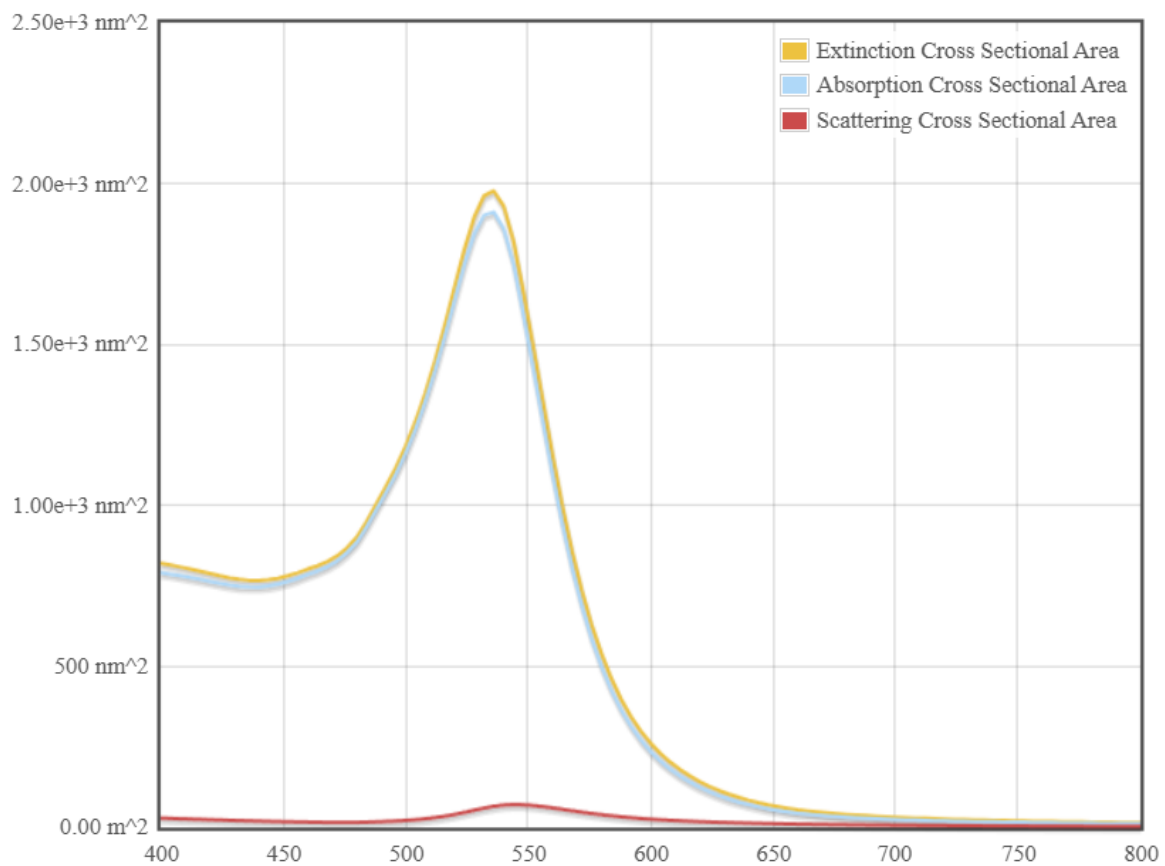
Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.

Core Type	Core RI	Core Diameter (nm)
Gold ▾		50
Shell Type	Shell RI	Shell Thickness (nm)
None ▾		0
Start λ (nm)	End λ (nm)	Medium RI
200	700	1.33

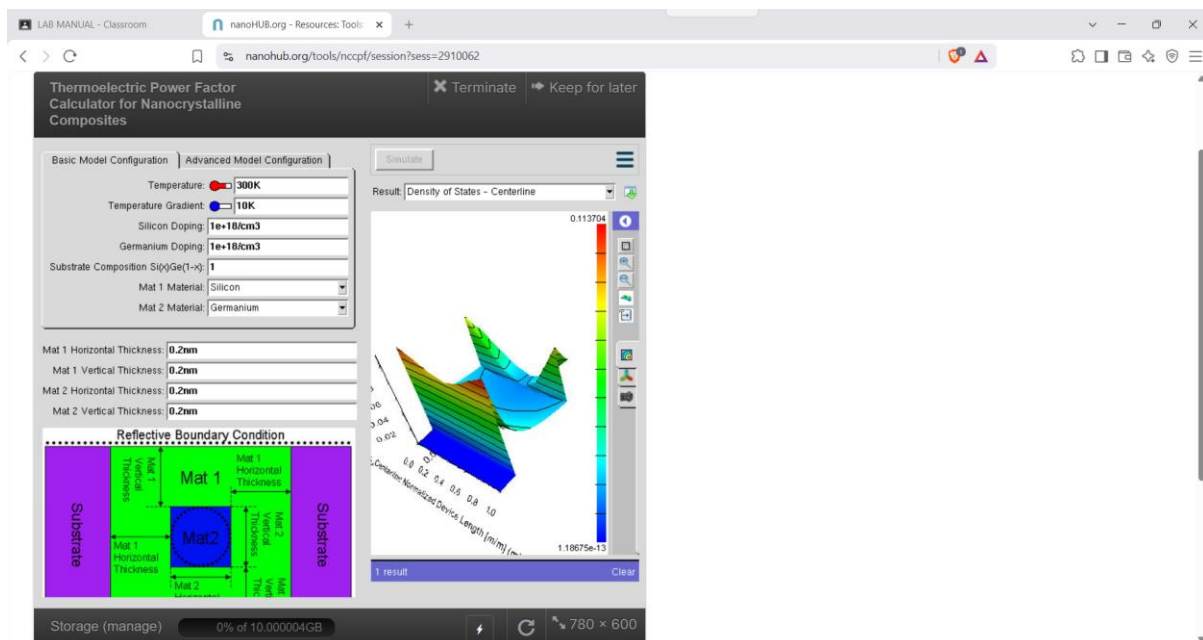


Case Study 3: Simulate a core-shell nanoparticle with a gold core (30 nm) and silica shell (10 nm)

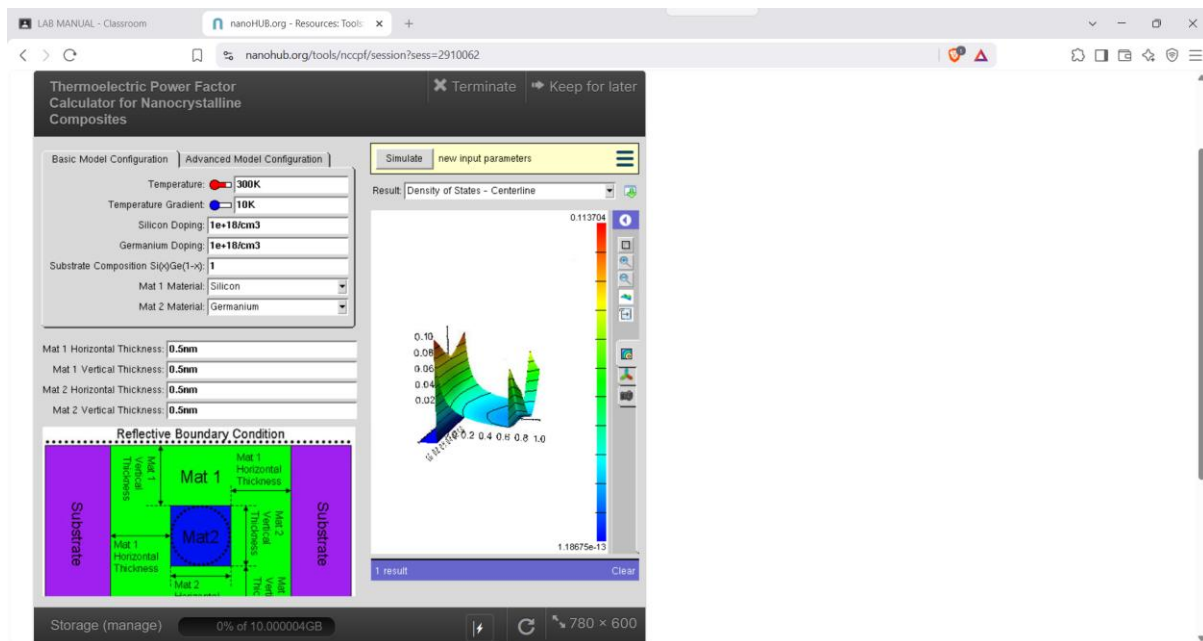
Core Type	Core RI	Core Diameter (nm)
Gold ▾		30
Shell Type	Shell RI	Shell Thickness (nm)
Silica ▾		10
Start λ (nm)	End λ (nm)	Medium RI
400	800	1.33



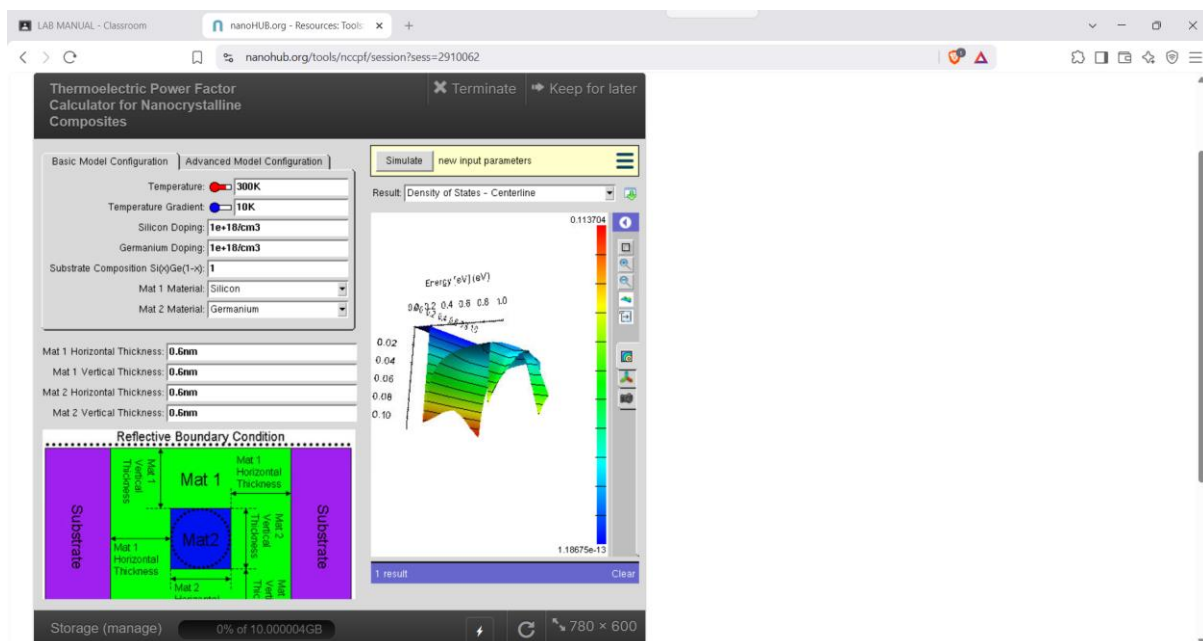
Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.



Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm

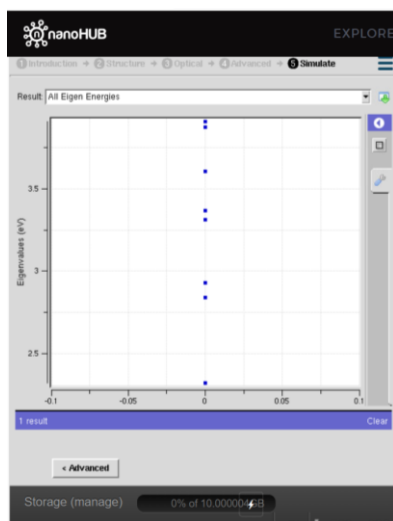
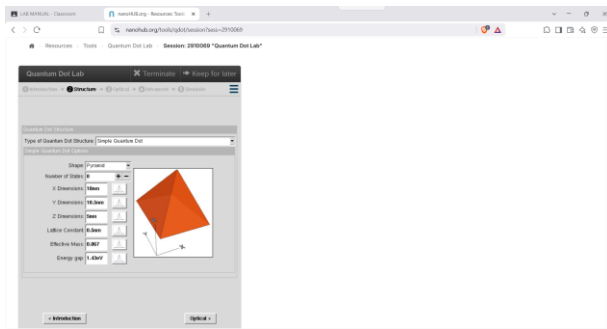


Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm

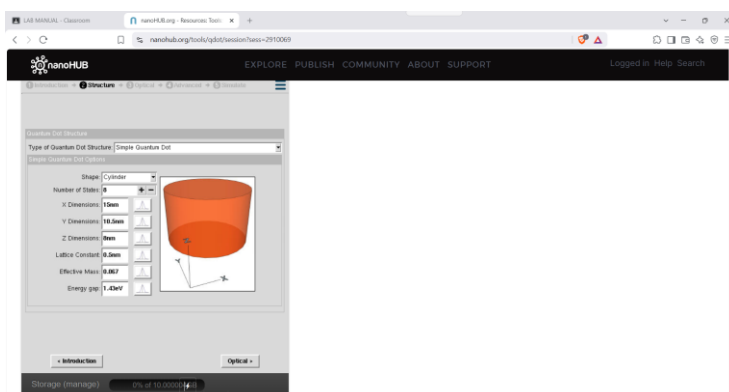


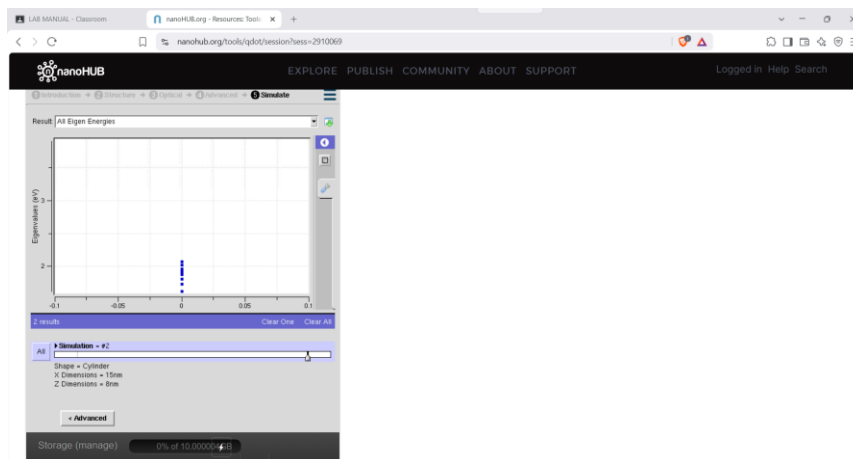
Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

Output:

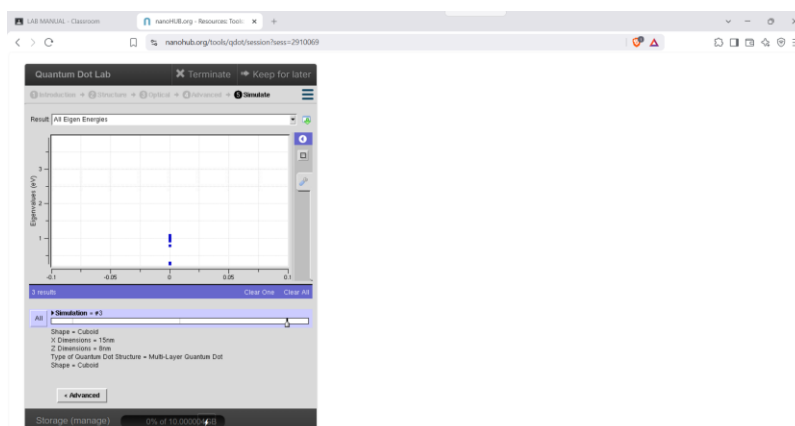
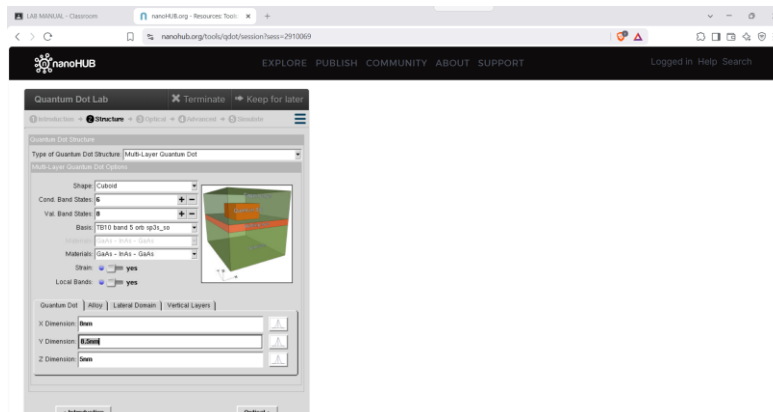


Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.





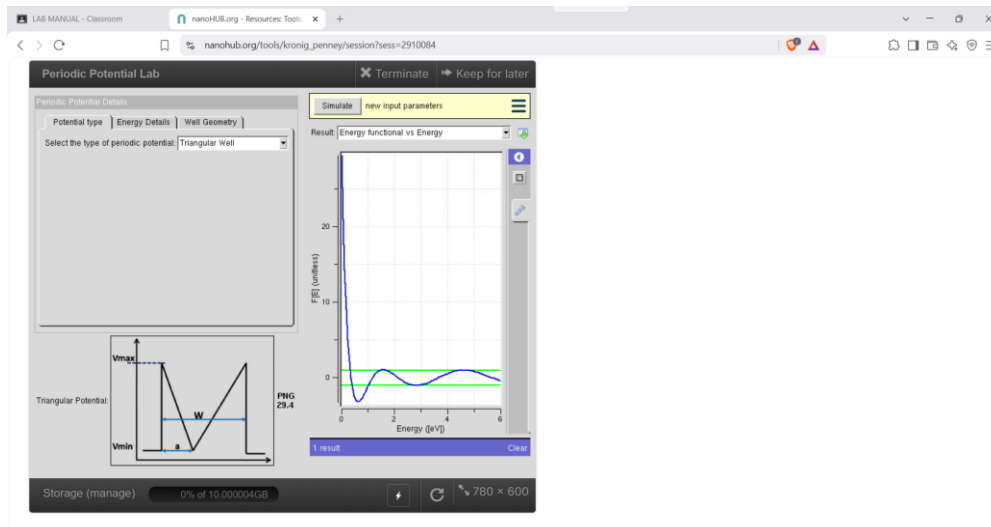
Case Study 2: Simulate the energy states of two layer cuboid quantum dots



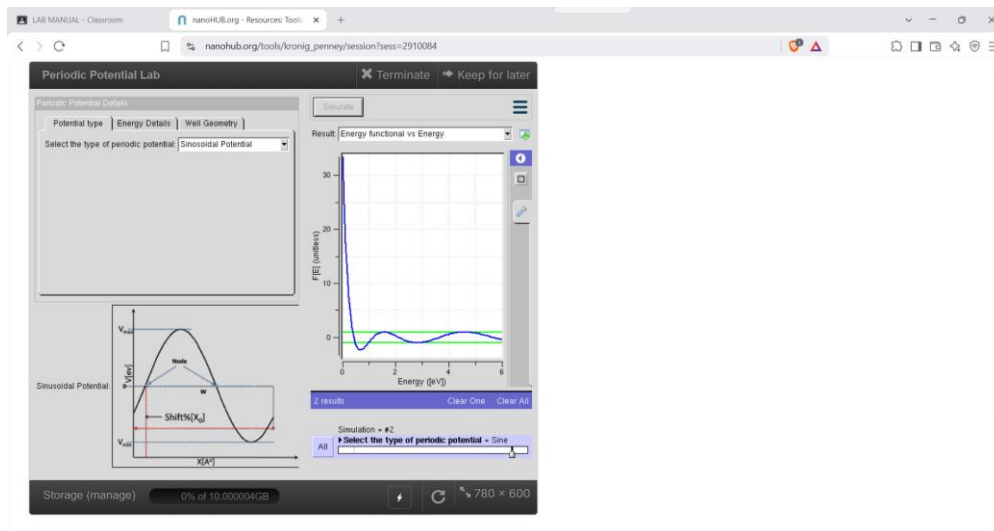
EXP. 4. KRONIG PENNEY MODEL



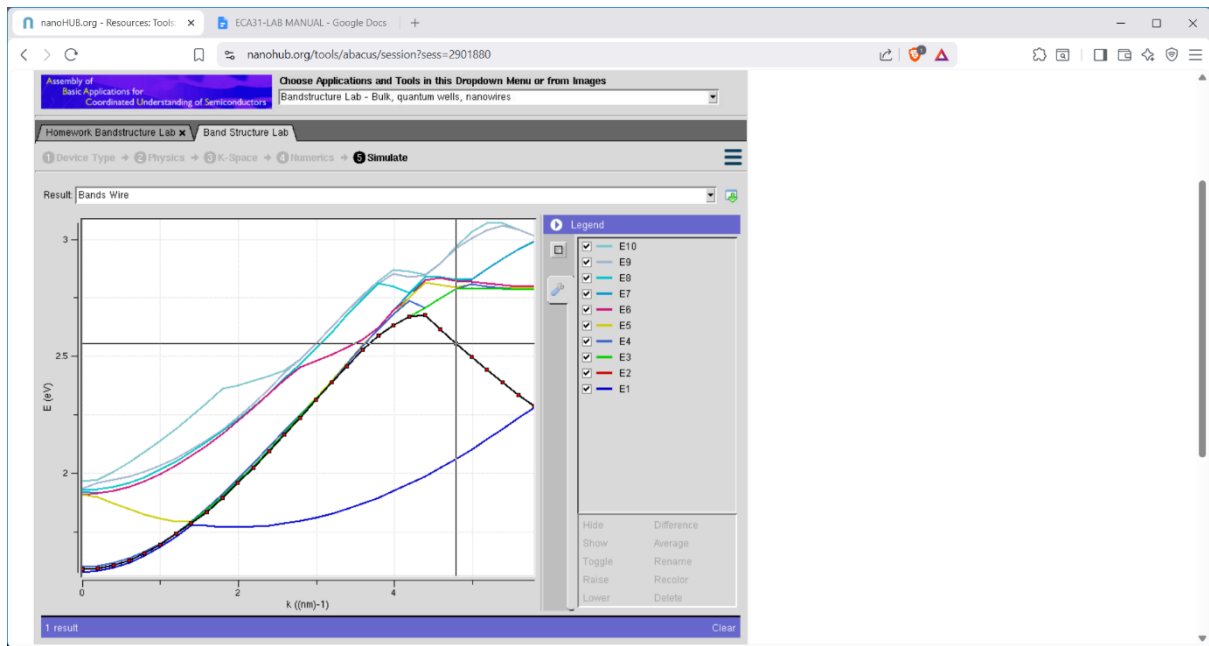
Case Study 1: Simulate the output for Triangular well



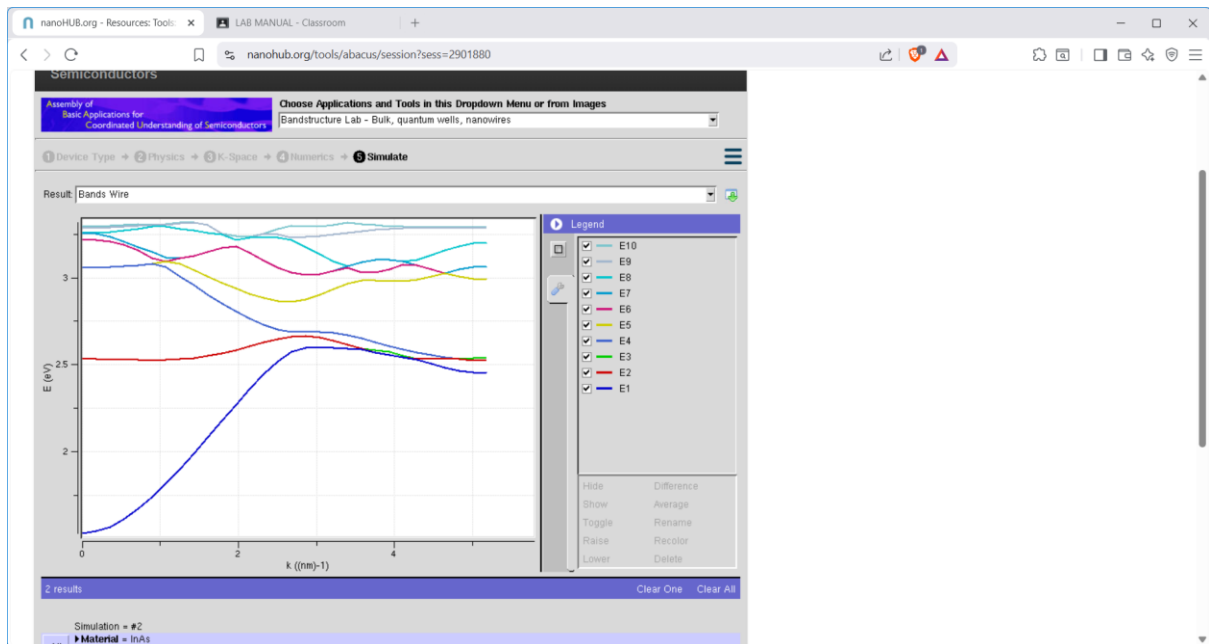
Case Study 2: Simulate the output for Sinusoidal Potential



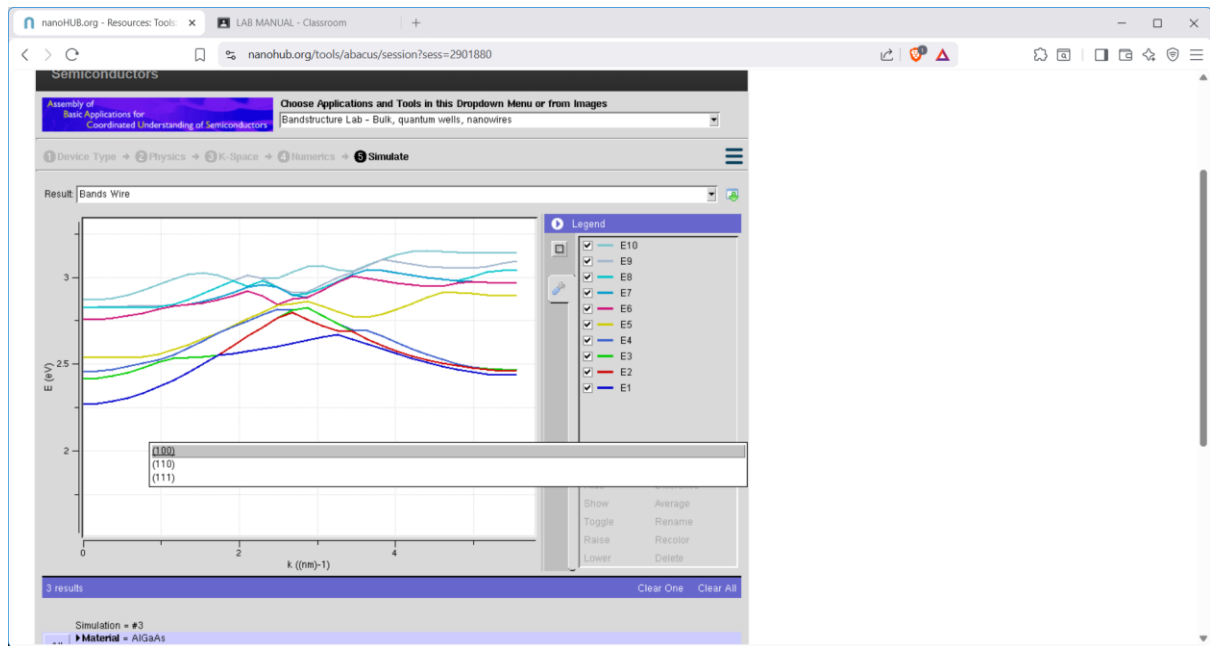
EXP. 5. BAND STRUCTURE SIMULATION FOR GRAPHENE AND CNTS USING NANO HUB



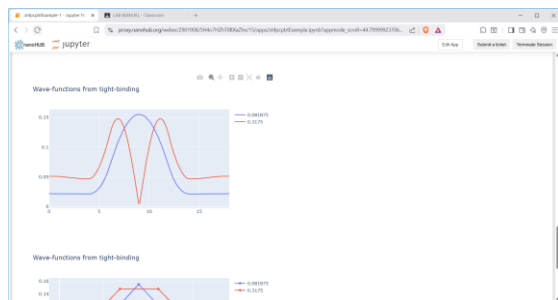
Case Study 1: Simulate the band-structure of InAs circular nanowire



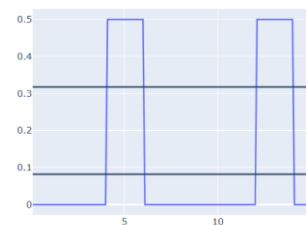
Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



EXP 6 DEMONSTRATION OF QUANTUM TUNNELING

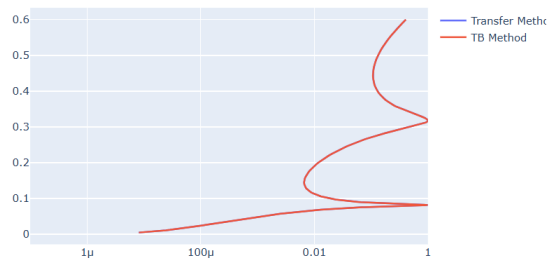


Potential Energy vs Distance

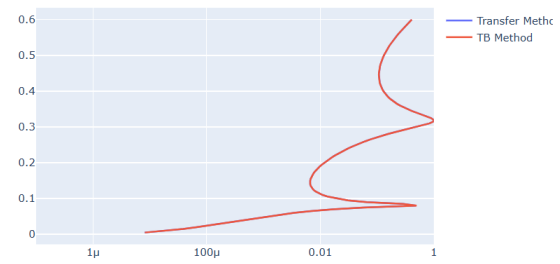


Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid

Transmission Coefficient Vs Energy on refined grid

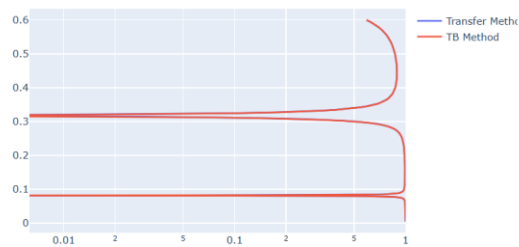


Transmission Coefficient Vs Energy on Homogeneous Grid

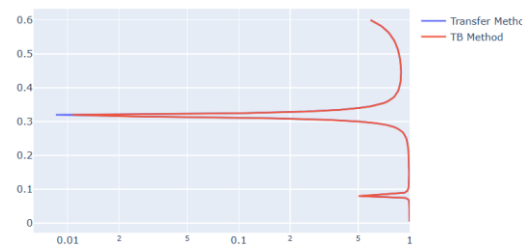


Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

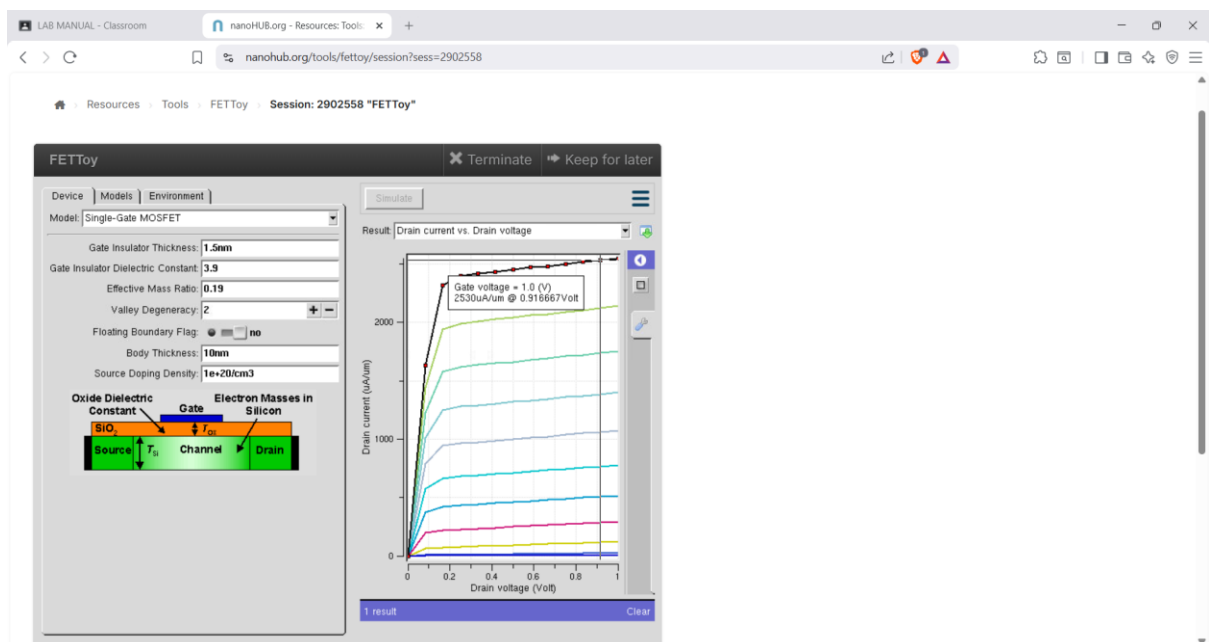
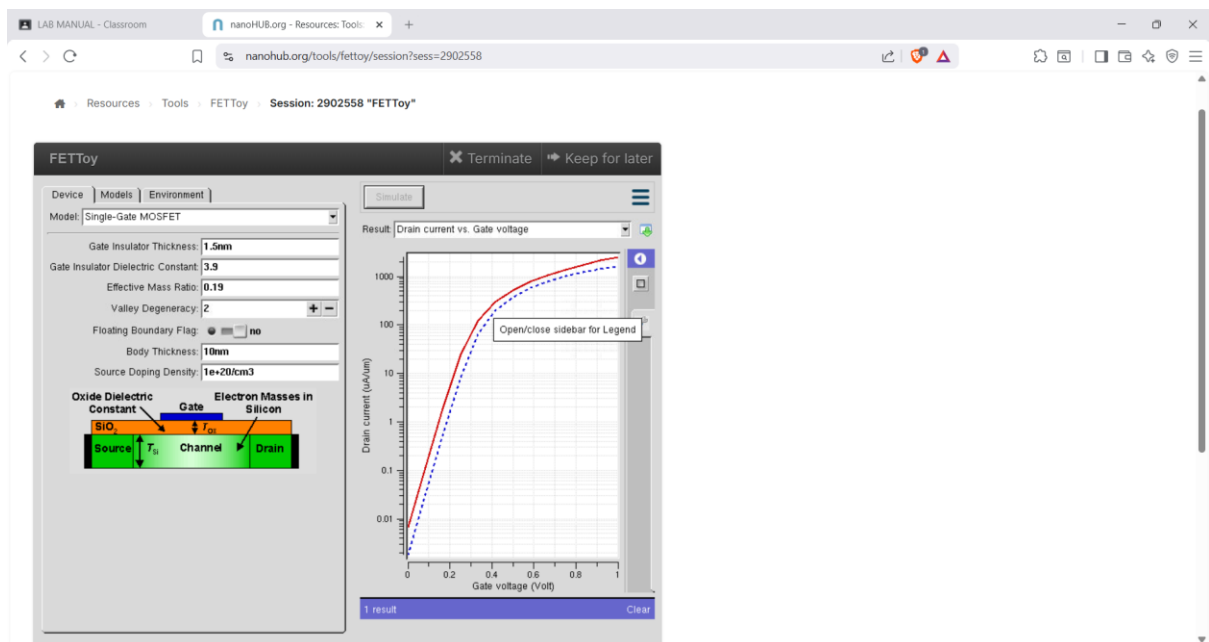
Reflection Coefficient Vs Energy on refined grid

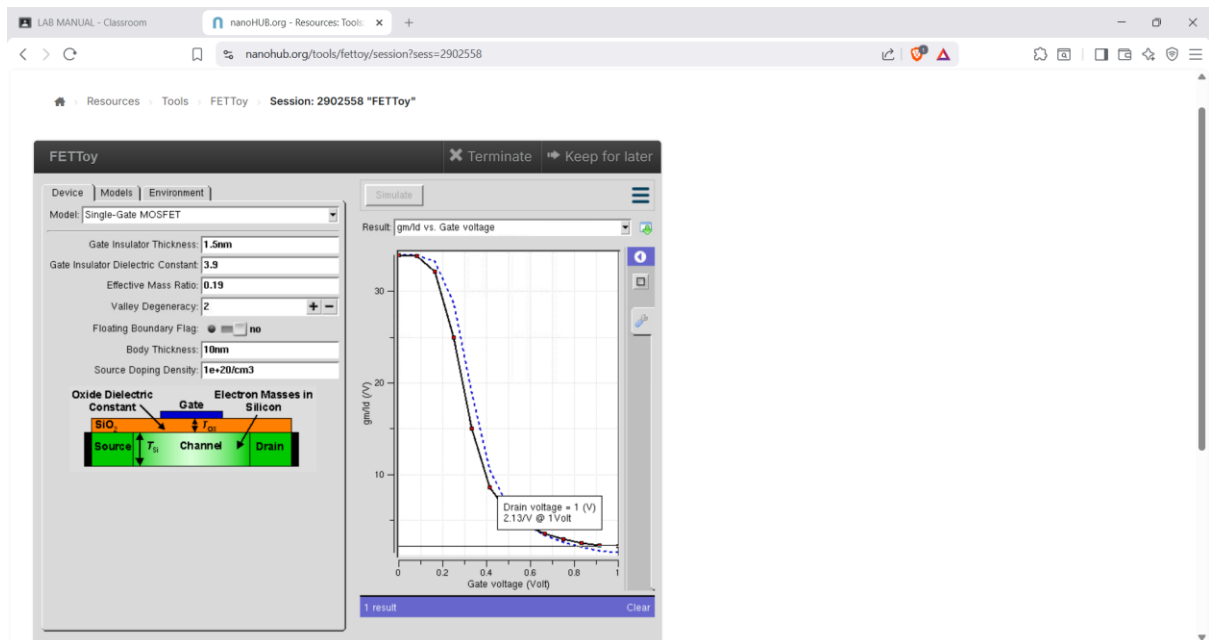


Reflection Coefficient Vs Energy on Homogeneous Grid



Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

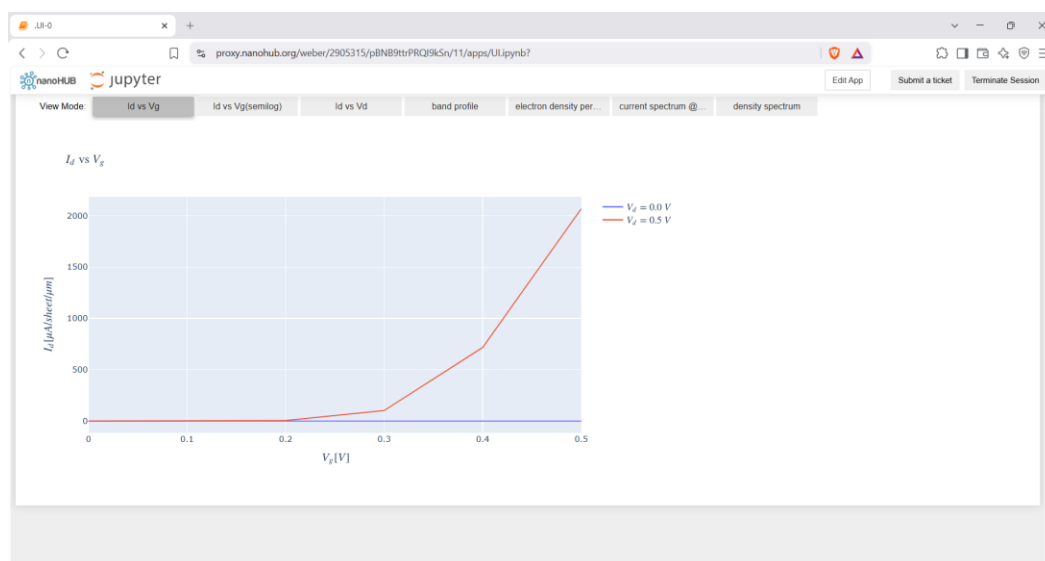


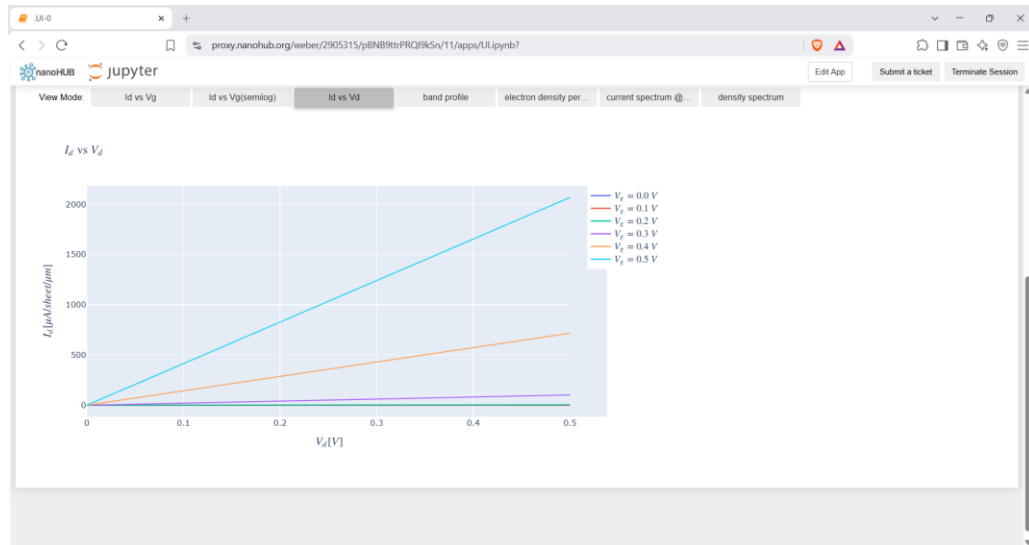


Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET

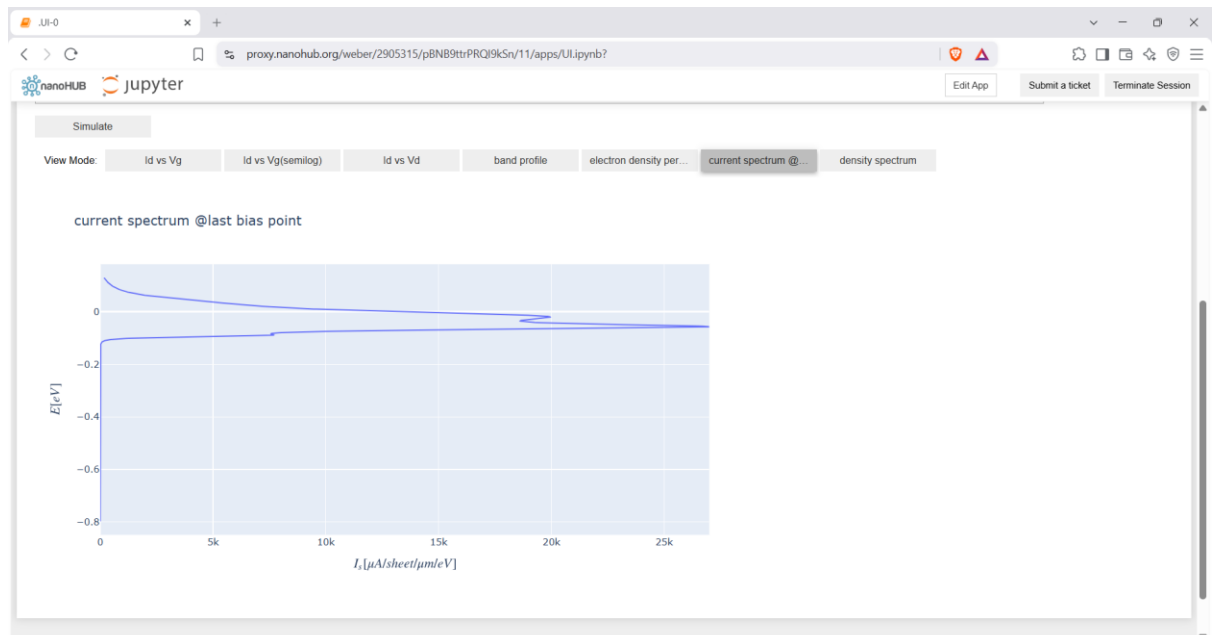
using NanoHub

(a) FINFET

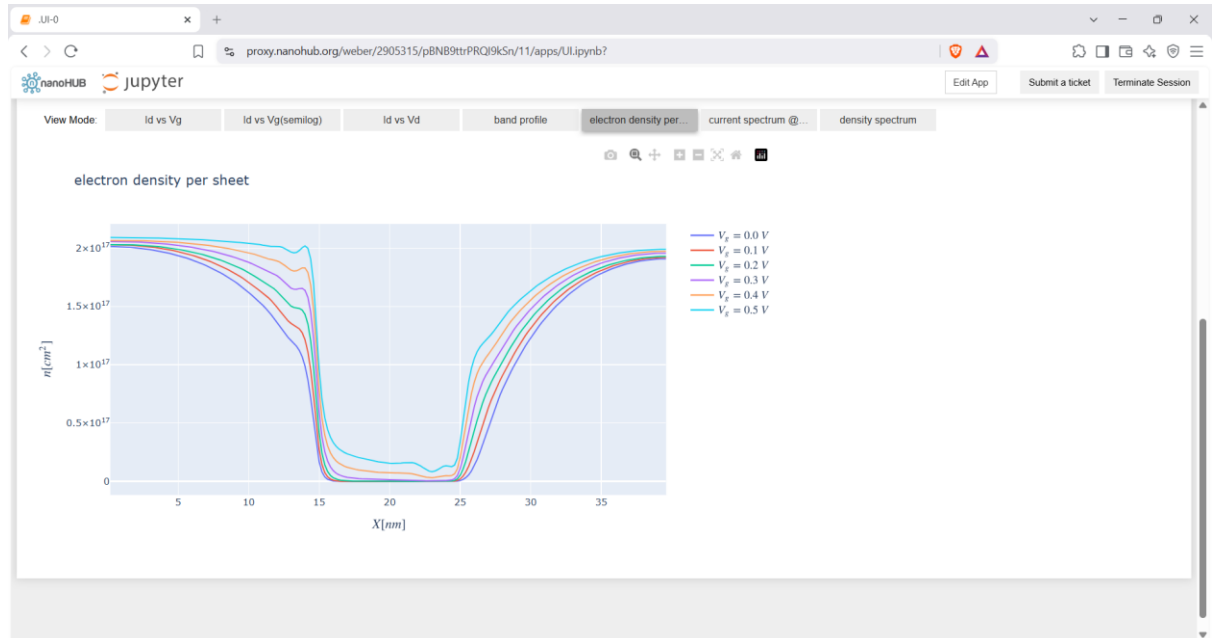




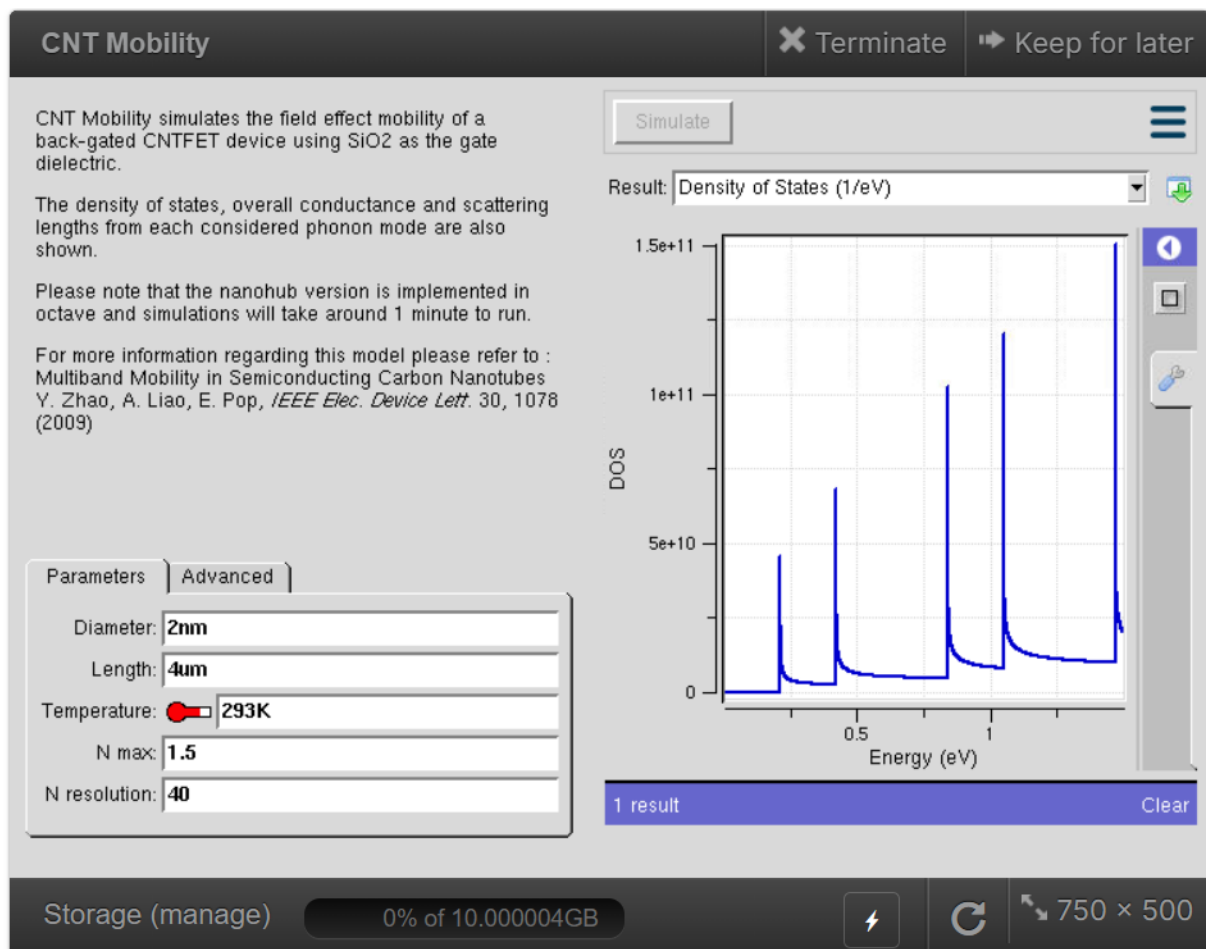
Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET. Analyze the performance for different Oxide layer thicknesses



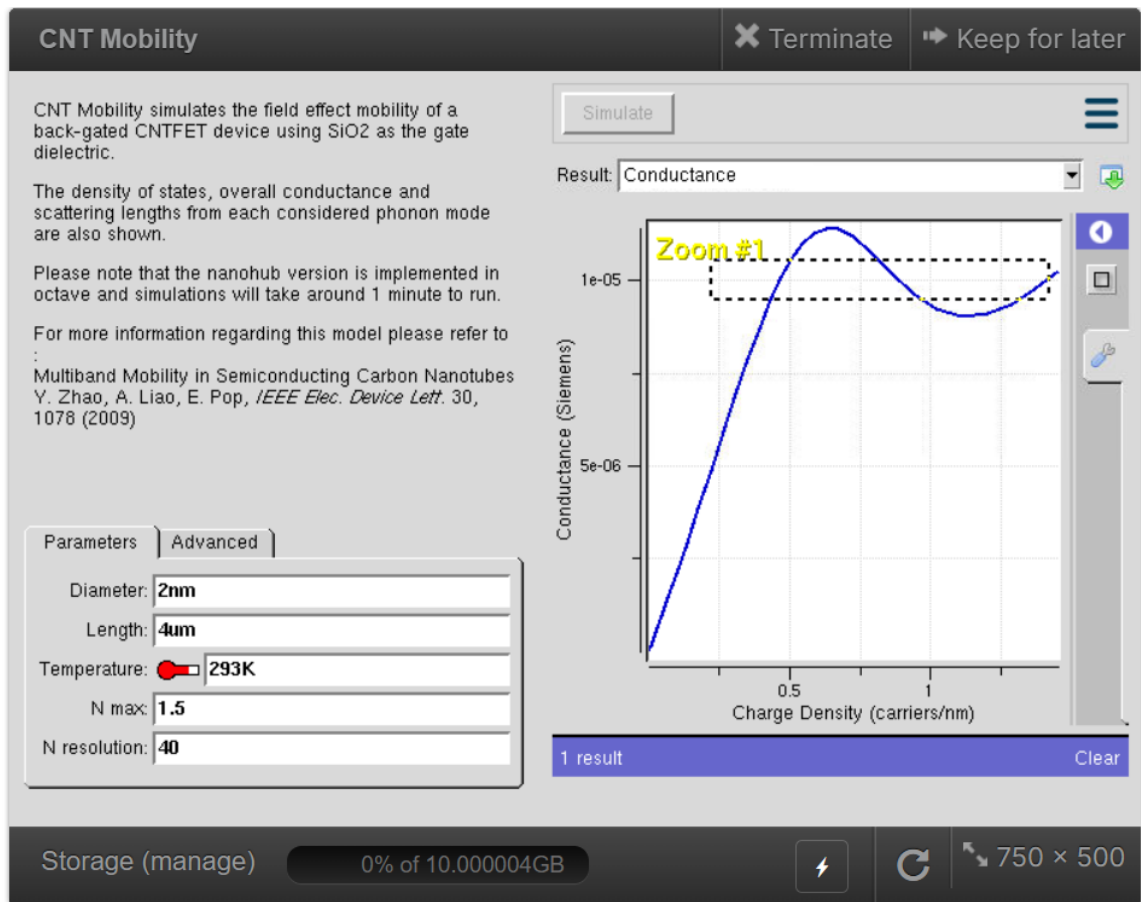
Case Study 2: Simulate & obtain the electron density per sheet characteristics of FinFET. Analyze its performance with different Gate region length



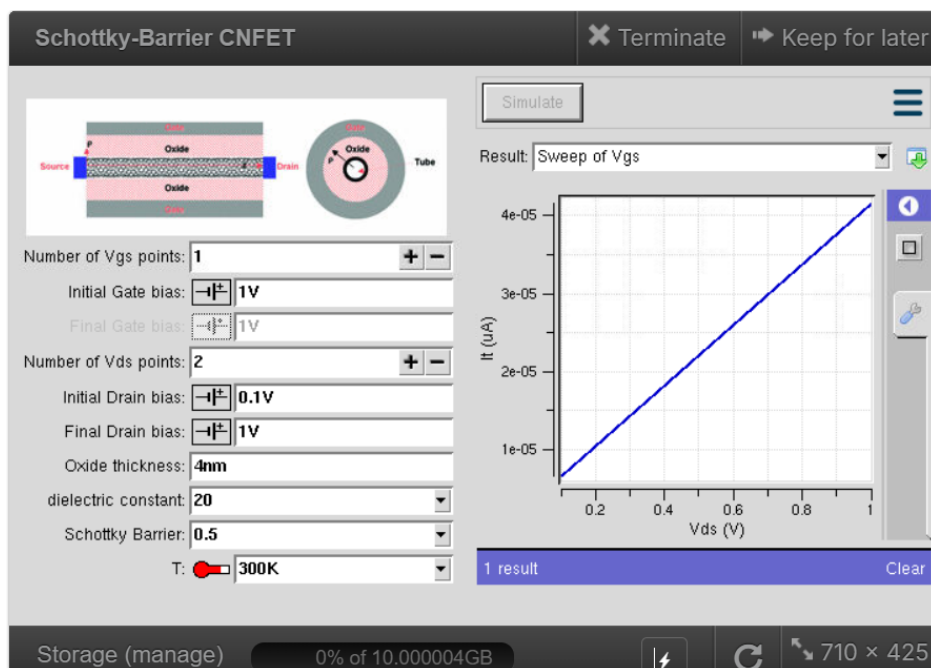
(b) CNTFET



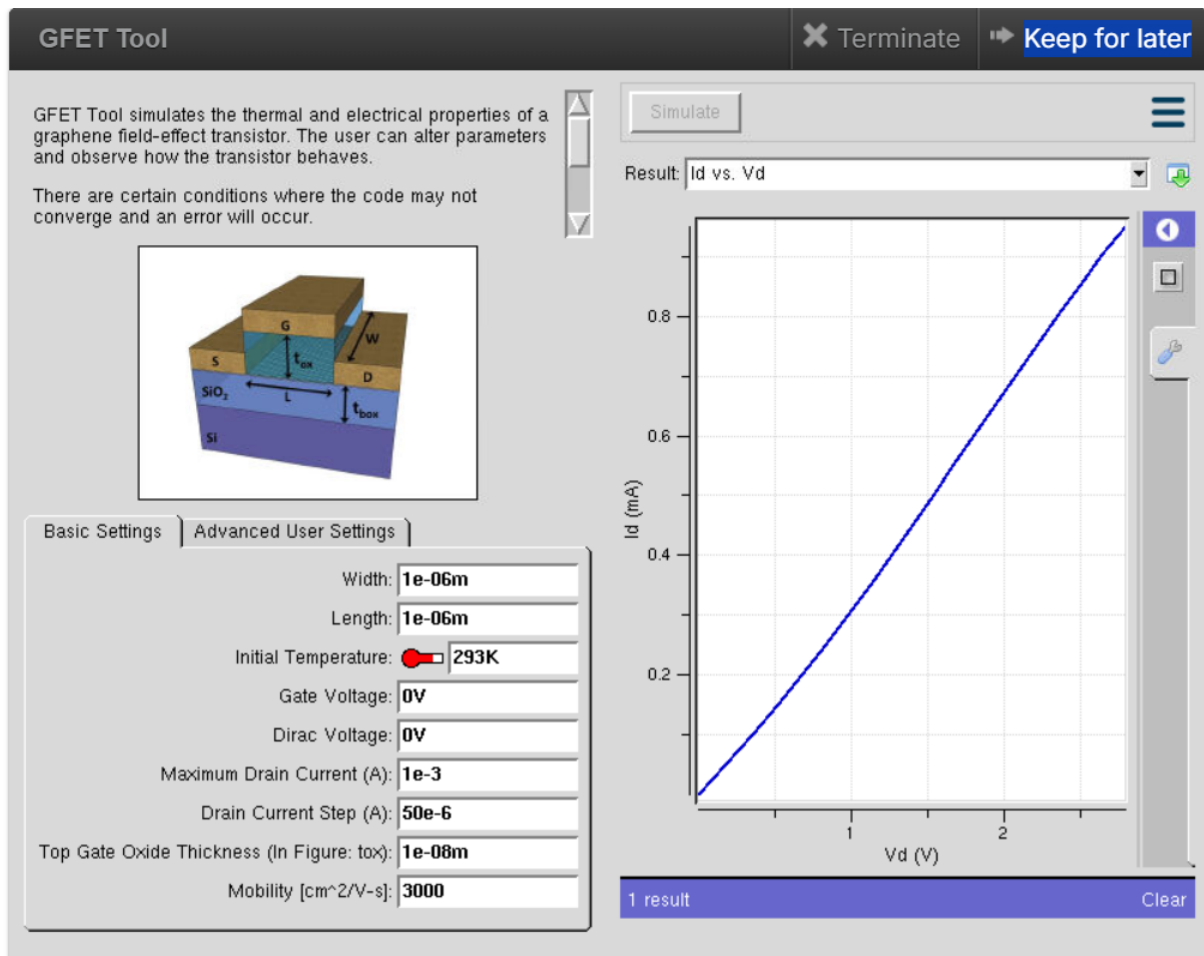
Density of states



Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness

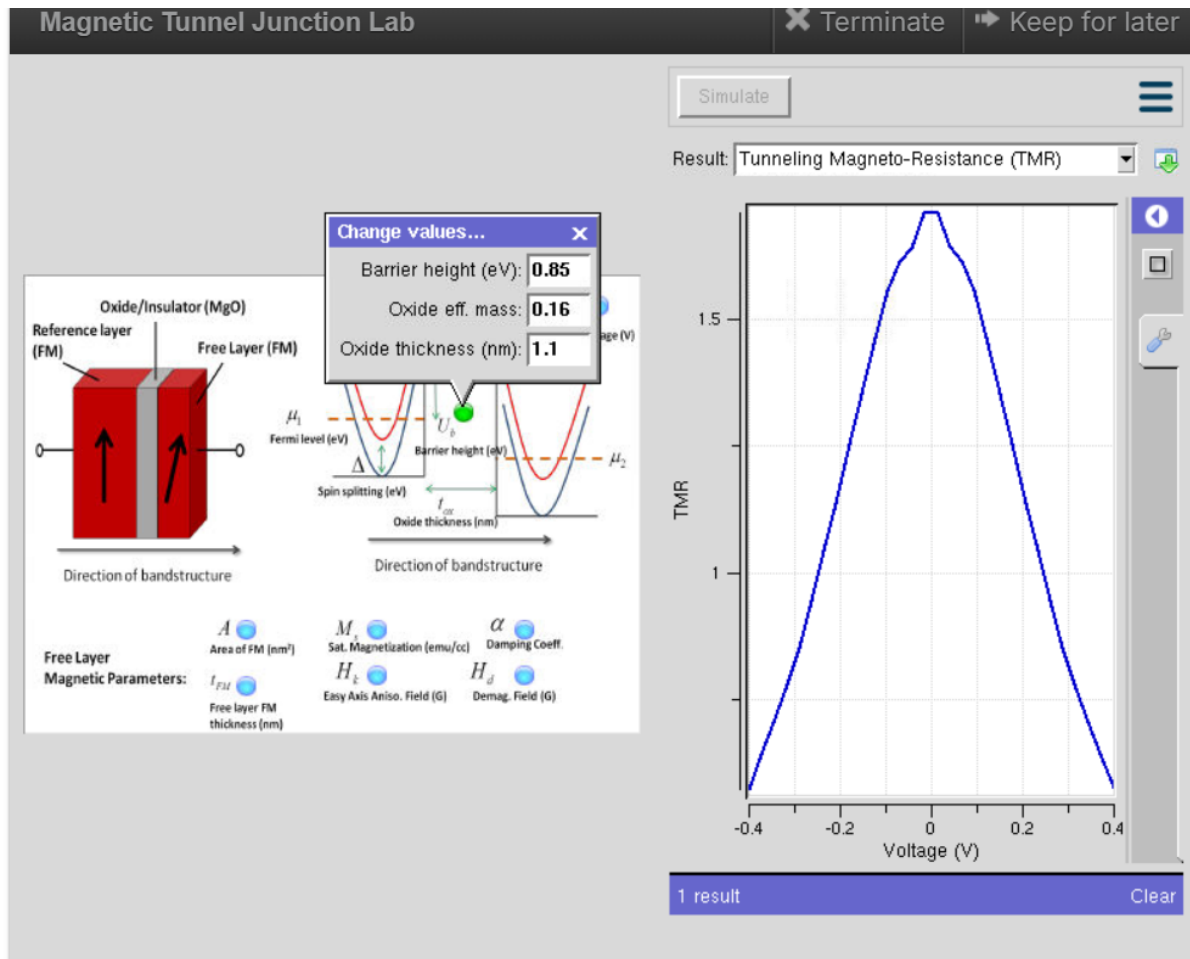


(c) GFET

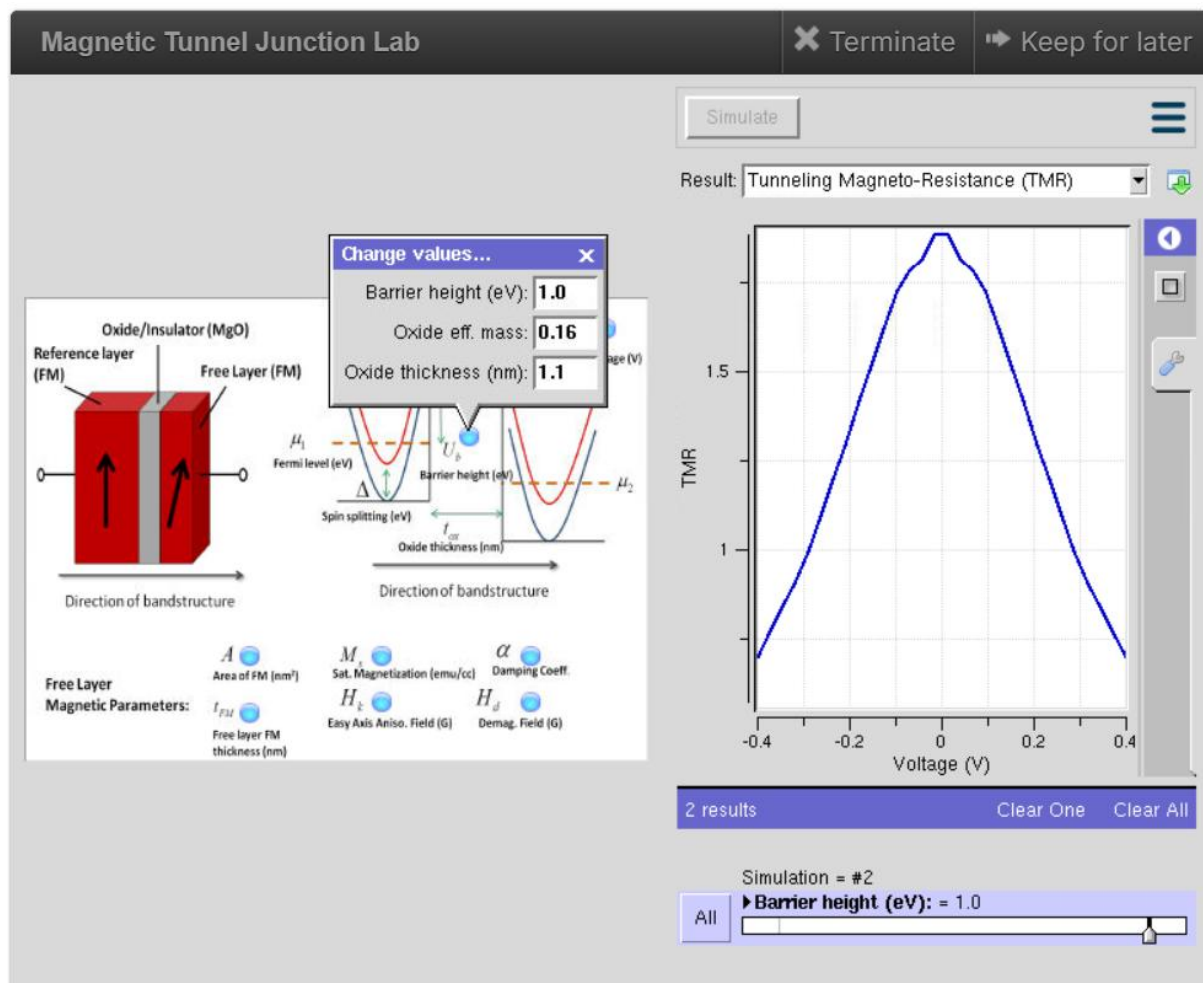


Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

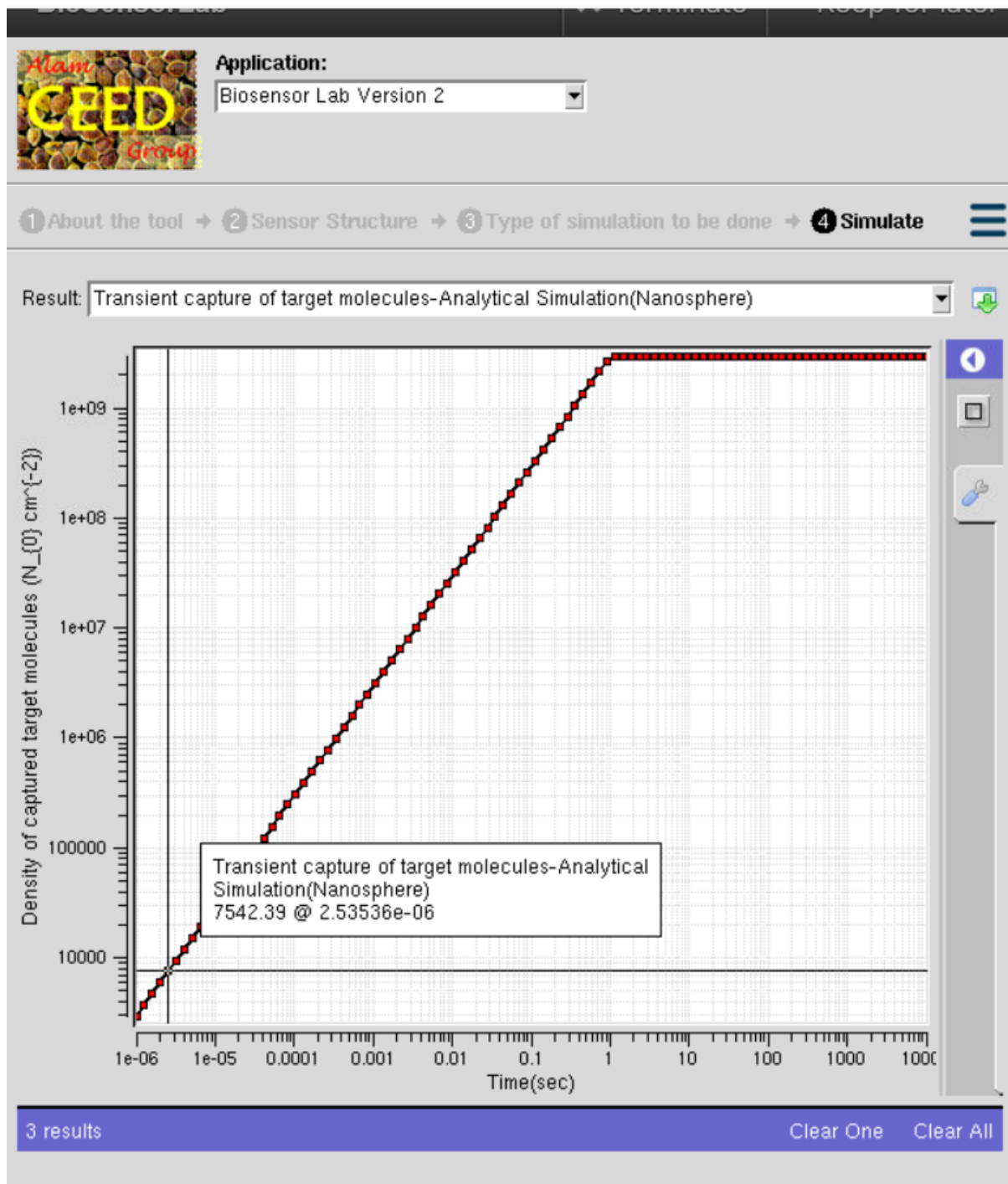
Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Case Study 2: Characteristics of TMR with barrier height of 1.0eV



Exp. 10. Biosensor and Bio-FET characteristics using NanoHub



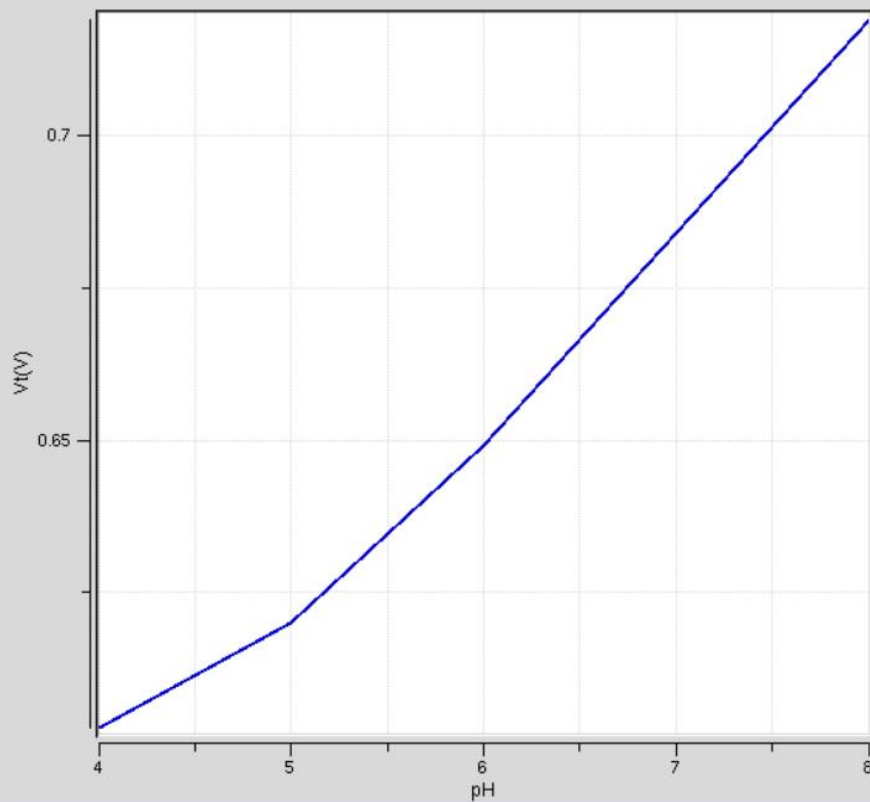


Application:

Biosensor Lab Version 2

1 About the tool → 2 Sensor Structure → 3 Type of simulation to be done → 4 **Simulate**

Result: Threshold voltage as a function of pH

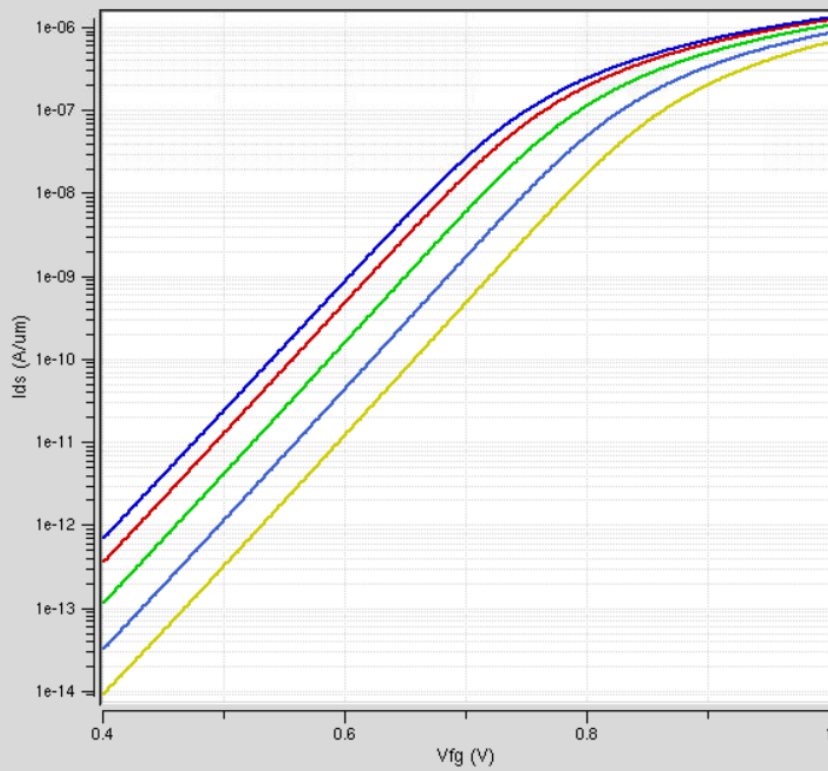


2 results

Clear One

Clear All

Result: Transfer characteristics(I_{ds} vs. V_{fg}) for different pH



2 results

Clear One

Clear All